

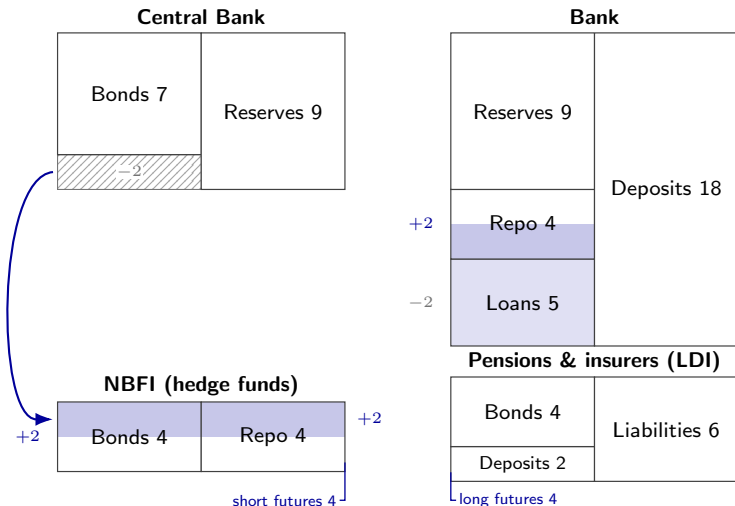
Bank to Non-Bank Lending and the Reallocation of Credit
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Supera

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Short summary

- ▷ **Research question:** Does bank lending to non-banks crowd out loans to firms?
- ▷ **Approach:** AnaCredit (2019–25) + a bank–NBFI portfolio model
- ▷ **Findings:**
 - Bank lending to NBFIs grew $\approx 80\%$ over 2019–25; to firms, $\approx 30\%$
 - Mostly repo to funds that hold securities, not to private credit
 - Driver: funds borrow to buy the extra bonds from QT and issuance
 - Table 6: when the NBFI share rises 1pp, the firm-loan share falls 0.56pp
 - Model: a 5% shift in bond supply $\Rightarrow$ total firm credit -6.3%
- ▷ **Convincing facts.** My discussion: how to interpret them, and the mechanisms

Short summary: the paper's mechanism



- ▶ **Banks dislike duration more than funds** $\Rightarrow$ funds buy the new bonds, on repo
- ▶ **Repo charged like a loan ($\approx 72\%$)**, **capital binds** $\Rightarrow$ repo crowds out firm loans
- ▶ **Experiment** (5% bond-supply shift): NBFI loans +3.6%, firm credit -6.3%

Plan for the discussion

- ① QT drains reserves too: repo can replace them without squeezing firm loans
- ② QT also drains deposits, and banks short of liquidity lend less
- ③ Until mid-2024, funds lent cash in repo: the flip can look like new lending

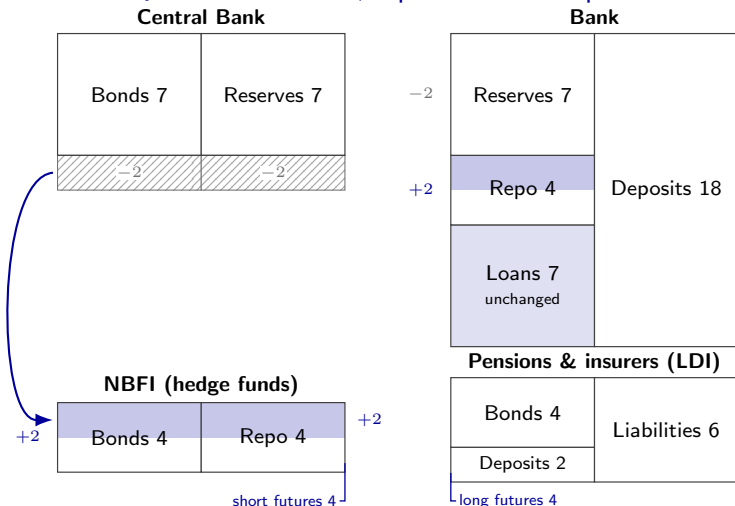
1a. Well-collateralised repo uses almost no risk-weighted capital

Equiv. risk weight	Paper's model	Basel rules
Reserves	0%	0%
Government-bond repo	$\approx 72\%^*$, pooled NBFI	$\approx 0\%$ if the bonds cover it

*My calculations: the paper's A5 capital formula at its pooled NBFI-loan inputs

- ▷ **At the paper's weights**, repo uses scarce capital and can squeeze firm loans
- ▷ **Paper's p.18**: such repos "have a risk weight of zero"
- ▷ **Why it matters**: repo drives growth; pooled weights can overcharge it
- ▷ **What repo does use**: balance-sheet space, in the leverage ratio

1b. When QT drains reserves, repo takes their space



- ▶ **Paper's experiment:** reserves fixed; the new bonds are financed on top (p.34)
- ▶ **With QT:** reserves become repo; principal leverage exposure is unchanged
- ▶ **Paper's p.3:** QT "frees up bank balance sheet capacity" for reverse repos
 - **Ask:** price repo by the rules, let reserves fall: what remains of -6.3% ?

1c. Table 6 is estimated in shares, read in euros

Estimated (eq. 4.6)

$$\Delta\left(\frac{\text{Firm loans}}{\text{Loans}+\text{securities}}\right) = \beta \cdot \Delta\left(\frac{\text{NBFI loans}}{\text{Loans}+\text{securities}}\right)$$
$$\beta = -0.56$$

Read as a level change, not a proportion change:

“... associated with a 0.56 percentage point decrease in firm loans” (p.19)

“the reallocation of credit away from firm lending” (p.19)

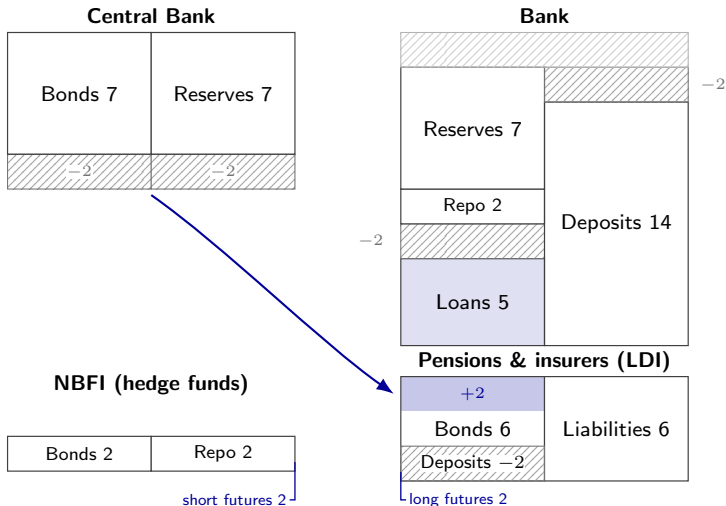
“largely at the expense of loans to non-financial firms” (p.3)

- ▶ **Why it matters:** if reserves aren't counted in loans + securities, swapping them for repo makes that total grow, so the firm-loan ratio falls while firm loans stay the same

1d. How much? A pure swap gives about a third of the -0.56

- ▶ **Euro-area banks:** firm loans $c \approx 17\%$ of loans + securities, NBFIs loans $n \approx 10\%$
- ▶ **A pure reserves-for-repo swap** then gives $-c/(1 - n) = -0.17/0.90 \approx -0.19$
- ▶ **So about a third of the paper's -0.56** can arise with no firm loan lost
 - **Ask:** Table 6 in euros on a fixed base; reserves and total book as outcomes

2a. When buyers pay out of deposits, bank liquidity falls



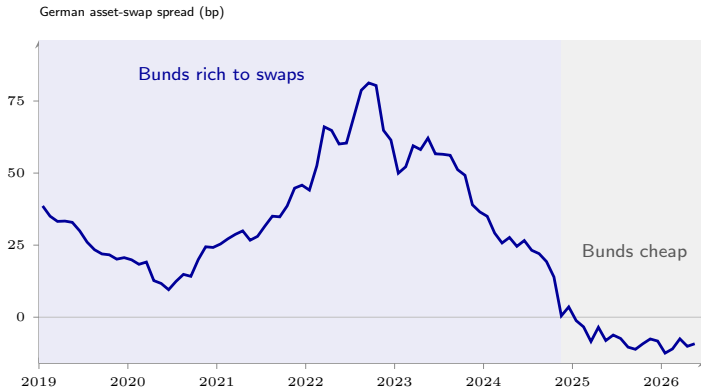
Liquidity coverage ratio: reserves $\geq \frac{1}{2}$ deposits: $7 = 7 \checkmark$

- ▶ **Chavaz, Patozi & Wazzi:** UK, 2021–24
- ▶ **Liquidity:** reserves count in full, lost deposits partly; the ratio can fall
- ▶ **Banks lend less:** “reducing lending shrinks the balance sheet”

2b. Both channels can be at work: model both, then split the effect

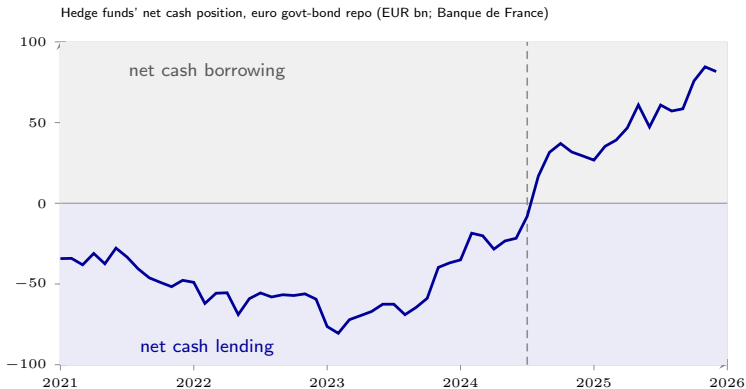
- ▷ **Paper's channel:** funds' demand for repo uses bank capital; firm loans give way
- ▷ **Liquidity channel:** reserves and deposits leave; banks short of liquidity lend less
- ▷ **Both fit the facts:** more repo and fewer firm loans at the same banks
- ▷ **Paper's model** has no liquidity constraint, so it cannot split the two
 - **Ask:** add a liquidity constraint; how much of the firm-credit fall is each?
 - **Test:** in Table 7, add banks' pre-QT liquidity buffer next to NBFI share

3a. Until late 2024, the euro basis was inverted: Bunds were rich



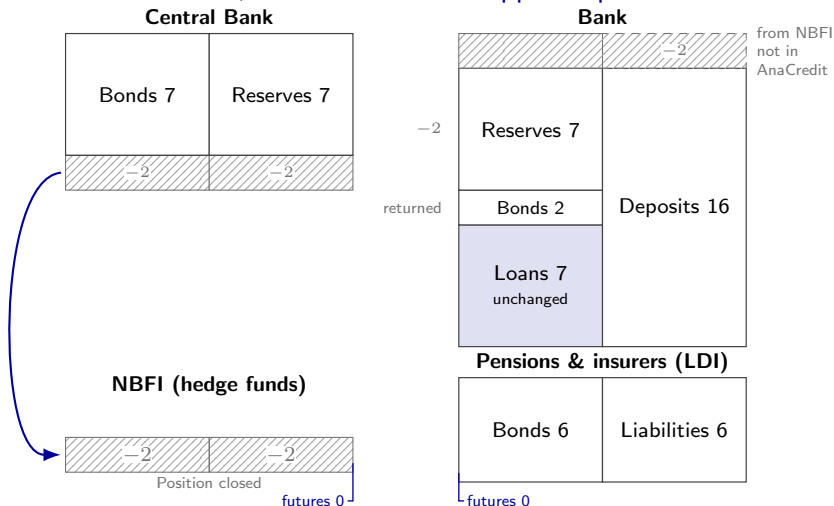
- ▷ Bunds traded rich to swaps: safe bonds were scarce
- ▷ Since 2025, Bunds trade cheap to swaps

3b. Until mid-2024, hedge funds lent cash in euro repo



- ▷ Net cash lenders until mid-2024; net borrowers since

3c. Until mid-2024, the funds held the opposite position



- ▶ **BdF:** covered hedge funds were net cash lenders in euro govt-bond repo
- ▶ **Illustration:** short Bunds, lend cash, long futures; pensions hold bonds
- ▶ **QT closes shorts:** reserves and the bank's repo liability fall together
- ▶ **Neither mechanism applies:** no new loan to funds; LCR does not fall

3d. In dollars, funds borrow bank cash, as in the paper's story

	Euro repo	Dollar repo
Basis	Bunds rich until 2025	usual: long cash, short futures
Hedge funds' net cash	lent until mid-2024	borrowed, $\approx$ \$1.8tn in 2025

- ▶ **Paper's fn. 8:** a marked rise in non-euro repo, "Treasury basis trades"
- ▶ **ECB:** euro-area banks intermediate it via US affiliates, on balance sheet
- ▶ **Open question:** does this balance-sheet use crowd out euro-area firm loans?
 - **Ask:** split repo by currency; relate dollar books to leverage headroom

Taking stock

- ▷ **The facts are convincing:** banks lend more to NBFIs, mostly in repo
- ▷ **What drives them is still open**
 - Repo can replace reserves at almost no capital cost
 - QT also drains deposits, and banks short of liquidity lend less
 - Until mid-2024, funds lent cash in euro repo rather than borrowed it
- ▷ **A more comprehensive model**
 - Repo priced by the rules, and reserves that fall with QT
 - A liquidity constraint next to the capital constraint
- ▷ **More data on the rest of the balance sheet should help**

Appendix: what the rules charge on repo

Rule	What the bank is charged on
Bilateral: comprehensive	Exposure net of haircut-adjusted collateral; capital on the residual.
Supervisory haircuts	Default ten-day haircuts start at 0.5% for short, high-grade sovereigns; scale for repo holding period and margining.
Eligible core participants	Zero haircut: 0%-RW sovereign, same currency, overnight or daily margin.
Qualifying CCP	Trade exposure carries a 2% risk weight.
Leverage ratio	Securities financing counts gross, with limited netting.
Liquidity ratio	Available, retained collateral may count as HQLA; Level 1 has 0% maturity inflow.

- ▷ **Risk-weighted capital:** ≈ 0 when haircuts cover exposure.
 - **Caveat:** Residual exposure remains if collateral or haircuts are insufficient.

Appendix: the paper's facts in the inverted regime

Paper's finding	Paper's reading	Our reading to test
Repo to hedge funds grows	Funds need bank cash to hold bonds	Closing shorts can shrink bank repo borrowing, unseen in AnaCredit
Repo rates rise with volume	Demand for bank cash rises	Rates rose from ≈ 40 bp below the DFR to about the DFR
ECB-held share falls: more repo, higher rates	More financing needed for that bond	Scarcity relief also raises repo rates
More NBFI lending, fewer firm loans	Bank cash diverted from firms	EGB repo: net cash lenders before mid-2024; split the estimates
Model: long funds absorb bonds	Banks move duration to funds	Short, hedged cash lenders lie outside this mechanism

- ▶ **Discriminating evidence:** repo-rate levels, net positions, swap spreads.
- ▶ **Our extension:** basis moves toward zero; net bank repo lending flips sign.

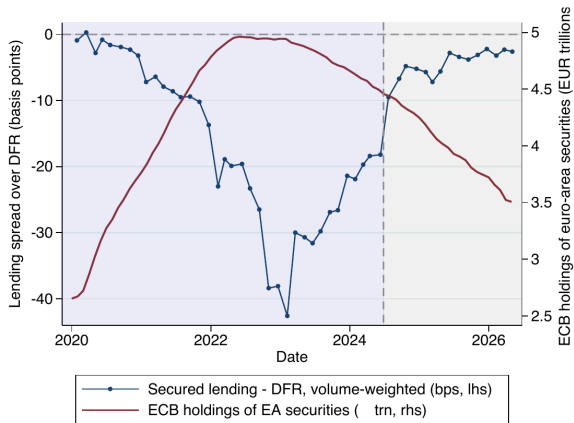
Appendix: same QT, another branch

Margin	Chavaz–Patozi–Wazzi	Li–Ma–Mendicino–Supera	Our joint reading
Lending	Bank-panel lending falls	Firm lending falls at NBFI-exposed banks	Agree on loan cuts
Exposed banks	Low liquidity; reserve-access frictions	Low reserves, deposit losers, low capital	Liquidity overlap; capital: the paper's channel only
Asset mix	Fewer loans, more liquid securities	More repo; bond holdings +€494bn	Holdings alone do not measure duration
Maturity	Average loan and asset maturity rise	Model: banks shun duration	A test of the premise; net of swaps
Exposure	Reserve-loss instrument	Initial NBFI lending share	Not tested against each other
Constraint	Rebuild liquidity; reserves leverage-exempt	Risk-weighted capital binds in model	Different mechanisms; final LCR response insignificant
Prices	Short fixes react more	Repo rates rise with volume	Repo levels also fit scarcity relief

▷ **Our reading:** liquidity loss can produce repo up and loans down.

→ **Ask:** horse race reserve losses and liquidity against NBFI exposure.

Appendix: the paper's Figure A.10



Below the DFR throughout QT

Appendix: excess liquidity fell mainly through TLTRO

- ▷ **Peak excess liquidity:** €4,748bn in November 2022.
- ▷ **Main driver through April 2024:** TLTRO III repayments
- ▷ **Also:** a gradual decline in APP holdings.
 - **Ask:** separate TLTRO repayments from bond runoff.

Appendix: the dollar side

Euro-area banks	ECB evidence
Dollar repo	Almost doubled since 2022 to €1.6tn (Nov. 2024).
Collateral	70% government bonds, of which 95% Treasuries (Jan.–Nov. 2024 average).
Dollar channel	Liquidity intermediated between US affiliates and non-banks.
Net dollar borrowing	Repos exceed reverse repos by $\approx$ €250bn (Nov. 2024).
Repo intermediation	More than doubled since 2021, driven by non-EA funds.
Secured-financing books	Large intermediaries typically keep balanced dollar/euro books.

▷ **Dollars:** a separate channel through US affiliates.

→ **Ask:** separate dollar and euro repo, and identify their funding.

Appendix: aggregate timing

Period	Hedge funds, net, EGB repo	Euro-area firm credit
2023–mid-2024	net cash lenders	stalled: -0.2% (Oct 2023)
mid-2024–2025	net cash borrowers	recovering: 1.7% (Dec 2024) 2.9% (Oct 2025)

- ▷ **Paper's identification is cross-sectional:** aggregate timing cannot reject it
→ **Ask:** split Tables 6–7 at mid-2024

Appendix: a maturity test of the two channels

	Capital: long loans → short repo (illustration)	Liquidity loss (UK evidence)
Firm loans	fall	fall
Banks add	short repo	liquid assets
Long-asset share	falls	rises

- ▶ **Paper's premise:** banks shun duration; repo for long loans shortens assets
- ▶ **UK:** QT raised short-fix spreads; fixes beyond five years “virtually unaffected”
- ▶ **UK bank panel:** average loan and asset maturity rise
- ▶ **Testable now:** AnaCredit records each loan's maturity and rate type
 - **Ask:** fixation, fixed-rate share, bond maturity — net of swaps